

Stakeholder → Feature Mapping

Prototype Reference Guide | Problem Statement 26099 – AI-Driven Standardization and Harmonization of Material Codes Across CPSEs

1. Master Stakeholder–Feature Mapping

Stakeholder	Primary Goal	Most Useful Features	Prototype Screens / Components to Build
MoPNG / Government	National standardization, governance, visibility and measurable savings	Common National Code; Material Standardization; Analytics Dashboard; Cost & Savings; Agentic AI; SAP/ERP Monitoring	National overview dashboard; CPSE comparison; savings summary; standardization progress; governance/audit view
CPSEs	Clean existing material master and map local codes to common national identities	AI Material Matching; Duplicate & Similarity Detection; Common National Code; Material Standardization; OCR; SAP/ERP Integration; SAP/ERP Monitoring	Upload/import page; AI matching results; material detail page; code mapping; OCR extraction; ERP sync/status
Procurement Teams	Find equivalent materials quickly and identify aggregation/savings opportunities	AI Material Matching; Common National Code; Cost & Savings; Voice-Based Search; Material Standardization; Analytics Dashboard	Material search; voice search; comparison screen; supplier/price comparison; demand aggregation; savings card
Engineering / Technical Experts	Validate whether AI-identified materials are technically identical/equivalent	Smart Approval; Duplicate & Similarity Detection; Material Standardization; OCR; AI Material Matching	Review queue; side-by-side specification comparison; confidence score; Green/Yellow/Red approval; approve/reject/modify
Inventory / Warehouse Teams	Improve material and stock visibility and reduce duplicate inventory identities	AI Material Matching; Duplicate Detection; Common National Code; Voice-Based Search; Analytics Dashboard; SAP/ERP Monitoring	Inventory search; national-code view; duplicate stock view; CPSE/location stock summary; alerts
IT / SAP / ERP Teams	Integrate, synchronize and continuously monitor the existing ERP ecosystem	SAP/ERP Integration; SAP/ERP Monitoring; Agentic AI; AI Material Matching; Common National Code	Integration setup; sync status; API/ERP monitoring; mismatch alerts; mapping logs; system health

2. Feature-by-Feature Prototype Requirements

Feature	Who Uses It	What the Prototype Should Do	Key UI Elements
AI Material Matching	CPSEs; Procurement; Engineering; IT	Compare descriptions, specifications, attributes and other material data; return likely matches with confidence.	Search/upload input; match results; similarity score; matched attributes; explanation; filters
Duplicate & Similarity Detection	CPSEs; Engineering; Inventory	Classify records as exact duplicate, near-duplicate, functionally equivalent/potential equivalent, or different.	Duplicate list; category/status badges; side-by-side comparison; confidence; filters
Common National Code	Government; CPSEs; Procurement; Inventory; IT	Recommend one national identity and maintain mapping from multiple legacy CPSE codes.	National code card; standard description; CPSE-code mapping table; mapping history
Material Analytics Dashboard	Government; Procurement; Inventory; CPSEs	Show material-master quality, duplicate trends, standardization progress and cross-CPSE insights.	KPI cards; charts; CPSE filters; category filters; trend charts; drill-down
Material Standardization	Government; CPSEs; Procurement; Engineering	Normalize descriptions, units and specifications and classify materials into a standard structure.	Before/after view; normalized attributes; category/classification; edit/approve controls

Feature	Who Uses It	What the Prototype Should Do	Key UI Elements
SAP/ERP Integration	CPSEs; IT/SAP	Exchange material data with existing ERP/SAP systems without replacing them.	Connection status; import/export; sync button; mapping status; integration logs
Voice-Based Search	Procurement; Inventory	Convert spoken material queries into searchable material requests.	Microphone button; transcript; search results; suggested materials
OCR Data Extraction	CPSEs; Engineering	Extract material codes, descriptions and technical attributes from PDFs, scans and images.	Upload/drag-drop; OCR preview; extracted fields; confidence; edit/confirm
Agentic AI	Government; CPSEs; IT/SAP	Coordinate matching, verification, migration, reporting and synchronization workflows.	Agent activity timeline; task status; recommendations; action log; automation controls
Smart Approval	Engineering; Material Master teams; Governance	Route AI recommendations by confidence: Green, Yellow, Red; let experts approve, reject or modify.	Review queue; Green/Yellow/Red badges; reason panel; approve/reject/modify
Cost & Savings	Government; Procurement	Compare material prices and estimate potential savings from standardization/demand aggregation.	Price comparison; demand aggregation; estimated savings; opportunity cards
SAP/ERP Monitoring	IT/SAP; CPSEs; Government	Regularly detect new/changed material records, mismatches and synchronization issues.	Monitoring dashboard; alerts; mismatch queue; last-sync time; change history

3. Recommended Prototype Flow

Step	User / System Action	Prototype Screen
1	CPSE uploads CSV/Excel, PDF, scan/image, or enters a material manually.	Data Upload / Material Input
2	System cleans and standardizes descriptions, units and attributes.	Data Processing / Standardization
3	AI extracts attributes and generates material similarity/match candidates.	AI Matching Results
4	System classifies candidates as duplicate / near-duplicate / equivalent / different.	Similarity & Duplicate View
5	AI recommends a Common National Material Code and maps legacy CPSE codes.	National Code Mapping
6	Smart Approval routes recommendations to technical/material experts.	Approval Queue
7	Approved mappings become part of the unified material master.	Unified Material Master
8	Dashboard displays material quality, inventory visibility, procurement opportunities and savings.	Analytics Dashboard
9	SAP/ERP integration synchronizes approved mappings; monitoring detects future mismatches.	Integration & Monitoring

4. Priority Screens for the Hackathon MVP

Priority	Screen	Purpose
P0 – Must Have	Unified Dashboard	Show total materials, duplicates, near-duplicates, standardized records, pending approvals and potential savings.
P0 – Must Have	AI Matching / Comparison	Show Material A vs Material B, similarity score, matched attributes and AI explanation.
P0 – Must Have	Smart Approval Queue	Green/Yellow/Red recommendations with Approve, Reject and Modify actions.
P0 – Must Have	Common National Code Mapping	Show national code recommendation and mapping of multiple CPSE legacy codes.
P1 – Strong Demo	Material Search + Voice	Search by text/voice and retrieve national code, equivalent materials and CPSE mappings.
P1 – Strong Demo	OCR Upload	Upload a sample PDF/image and show extracted material fields.
P1 – Strong Demo	Cost & Savings	Show price differences, combined demand and estimated savings opportunity.
P1 – Strong Demo	SAP/ERP Monitoring	Show mock integration status, last synchronization and mismatch alerts.

5. Key Design Principle

AI recommends → Expert validates → Governance approves → National code is mapped → ERP synchronizes → Monitoring continuously checks for new mismatches.

Prototype note: For the hackathon, SAP/ERP integration and monitoring can be demonstrated with mocked APIs/sample data while clearly presenting the production integration architecture.